

|   |         |                                                                                                                                                                                                                                                                         |                |     |
|---|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|
| 2 | (a) (i) | sum of kinetic and potential energies of the molecules<br>reference to random distribution                                                                                                                                                                              | M1<br>A1       | [2] |
|   | (ii)    | for ideal gas, no intermolecular forces<br>so no potential energy (only kinetic)                                                                                                                                                                                        | M1<br>A1       | [2] |
|   | (b) (i) | <i>either</i> change in kinetic energy $= \frac{3}{2} \times 1.38 \times 10^{-23} \times 1.0 \times 6.02 \times 10^{23} \times 180$<br>$= 2240 \text{ J}$<br><br><i>or</i> $R = kN_A$<br>energy $= \frac{3}{2} \times 1.0 \times 8.31 \times 180$<br>$= 2240 \text{ J}$ | C1<br>A1       | [2] |
|   | (ii)    | increase in internal energy = heat supplied + work done on system<br>$2240 = \text{energy supplied} - 1500$<br>energy supplied = 3740 J                                                                                                                                 | B1<br>C1<br>A1 | [3] |